Knowledge Graphs of Bank Volatility in the COVID-19 Crisis *

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1 Introduction

A knowledge graph represents entities as vertices and their relationships as edges. This project examines whether the volatilities of banks listed in the United States were more closely connected during the COVID-19 crisis than outside it. Chen and Zhang (2024) applied knowledge graphs to assess systemic risk in U.S. banking. They estimated a sparse precision matrix over the leading financial firms and drew an edge wherever an off-diagonal entry exceeded a cut-off, which they set separately for each graph. They compared the graph during the 2007–2009 financial crisis to the graph outside of that period. Regarding the liquidity discount index (Chen et al., 2016), they found that firms were more closely interconnected during the crisis, in line with the common view that firms are more connected during crises. Regarding volatility, they reported that the non-crisis period produced a more complex network than the crisis period.

This project extends their framework to the pandemic episode of 2020. We follow 70 banks listed in the United States at daily frequency from 2017 to 2021, and compare the graph inside the pandemic window with the graph outside it at a cut-off common to both. The crisis graph is denser only at small cut-offs on the absolute partial correlation. At 0.035, the 25 largest banks share 244 of the 300 possible edges inside the crisis against 175 outside it, whereas at 0.050 the crisis graph keeps 8 edges against 84, and the non-crisis edges keep a split into a North American and a European group.

Section 2 describes the data. Section 3 applies the embeddings and Section 4 estimates the knowledge graph. Section 5 concludes.[1]

2 Data

The universe is 70 bank stocks listed in the United States, taken from the ranking by market capitalization published by *Stock Analysis*. Daily adjusted closing prices were collected from May 2017 to December 2021. We define the volatility $\sigma_{i,t}$ of bank i on day t as the annualized standard deviation of its daily log returns over the trailing 252 trading days (1 year), so the study window runs from June 2018 to December 2021. The crisis window is defined as December 2019 to May 2020, which contains the event day of 20 January 2020 in Liu et al. (2020), and the non-crisis window is the remainder of the study period. Because the window spans 1 year, the subsamples split the days on which volatility is measured rather than the returns: crisis volatility draws on average 25% of its returns from the crisis window, and non-crisis volatility still contains crisis returns until May 2021. Table 1 states the windows, and Table 2 reports summary statistics.

3 Embedding

Let $z_{i,t} = s_i^{-1}(\sigma_{i,t} - \bar{\sigma}_i)$ be the standardized volatility, where $\bar{\sigma}_i$ and s_i are the sample mean and standard deviation of $\sigma_{i,t}$ over the study window in this section and over each subsample in Section 4. Three embeddings map the standardized volatility series of each bank to a point

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[1]Code and replication data are available at *GitHub*.

Period	Start	End
Full study	June 2018	December 2021
Crisis	December 2019	May 2020
Non-crisis	June 2018	November 2019
	June 2020	December 2021

Table 1: Definition of the subperiods.

Year	Mean (%)	Median (%)	SD (%)
2017	+17.5	+11.9	22.3
2018	−22.0	0.0	27.3
2019	+18.2	+30.4	29.0
2020	−12.7	−8.5	62.0
2021	+18.5	+14.9	31.2

Table 2: Daily log returns of the 70 banks pooled by calendar year, annualized.

in two dimensions, namely locally linear embedding (LLE), spectral embedding and principal component analysis (PCA). The first two are nonlinear methods that preserve local neighborhoods, and the third is the linear baseline. A k-means partition of each embedding with $k = 5$ colors its points, an auxiliary cross-check that Chen and Zhang (2024) also use.

Figure 1 shows the three embeddings of all 70 banks over the study window. LLE maps most banks into a dense mass near the origin and leaves a few apart. Spectral embedding places the banks along an arch, and PCA spreads them across both axes.

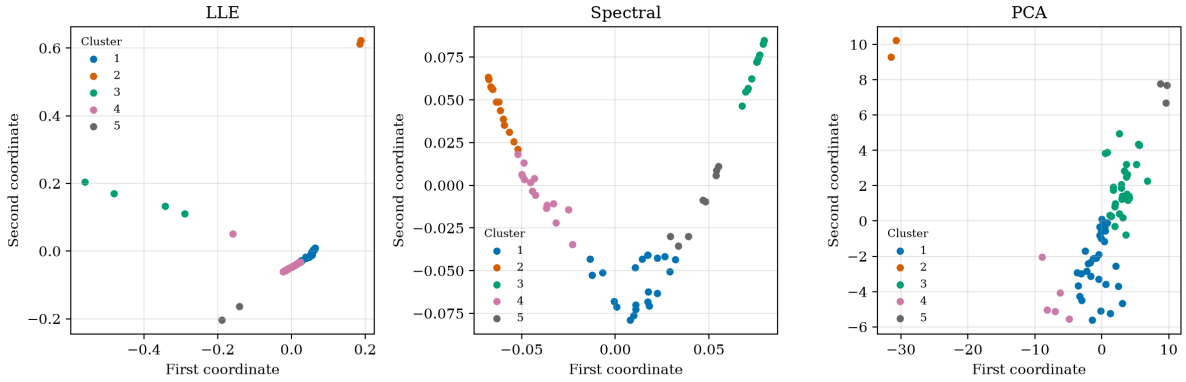


Figure 1: Embeddings of the standardized volatility of all 70 banks over the study window, colored by a k-means partition with $k = 5$.

4 Graphical Lasso

Next, we estimate the precision matrix of the 25 largest banks by the graphical lasso (Friedman et al., 2008), the ℓ_1-penalized Gaussian maximum likelihood estimator. Let S be the sample correlation matrix of their $z_{i,t}$ over a subsample. The graphical lasso estimates

$$\hat{\Theta} = \arg\min_{\Theta \succ 0} \left\{ \operatorname{tr}(S\Theta) - \log\det\Theta + \alpha \sum_{i \neq j} |\Theta_{ij}| \right\}, \tag{1}$$

where the penalty α is chosen by 3-fold cross-validation of the Gaussian log-likelihood. Under a Gaussian graphical model, $\Theta_{ij} = 0$ if and only if banks i and j are conditionally independent given the other 23 banks. The entries of $\hat{\Theta}$ are not comparable across two separate fits, since each carries the scale of its own diagonal. The edge weight is therefore the partial correlation

$$\hat{\rho}_{ij} = -\frac{\hat{\Theta}_{ij}}{\sqrt{\hat{\Theta}_{ii}\hat{\Theta}_{jj}}} \in [-1, 1], \qquad E(c) = \#\left\{(i,j) : i < j,\ |\hat{\rho}_{ij}| > c\right\}, \tag{2}$$

where c is a cut-off. In the spring layout, edge width is proportional to $|\hat{\rho}_{ij}|$ and vertex color is the k-means label of the PCA embedding of the 25 banks within the subsample.

Table 3 reports the selected penalty and partial-correlation statistics. The penalty is $\alpha = 0.600$ inside the crisis and $\alpha = 0.599$ outside it, so the two fits are penalized to the same degree. In both subsamples, however, the cross-validated log-likelihood rises as α decreases to that value, and every smaller penalty on the grid yields a non-finite score on at least one fold, so α lies at the boundary of the searched grid rather than at an interior maximum. At this penalty, $\hat{\Theta}$ has no zero off-diagonal entry inside the crisis and 21 of 300 outside it, so the edges are set by the cut-off c. The two subsamples have nearly the same median absolute partial correlation, 0.0397 versus 0.0386. However, the standard deviation outside the crisis is more than three times as large as during the crisis.

	Crisis	Non-crisis
Selected penalty α	0.600	0.599
Median $\lvert\hat{\rho}_{ij}\rvert$	0.0397	0.0386
Standard deviation of $\lvert\hat{\rho}_{ij}\rvert$	0.0061	0.0199

Table 3: Selected penalty and absolute partial correlations of the 25 largest banks.

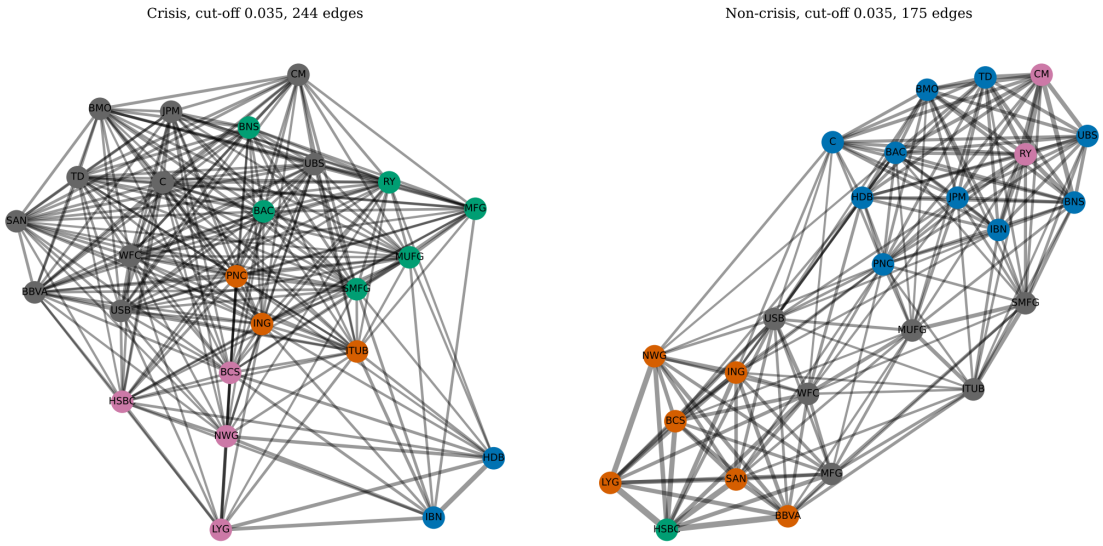


Figure 2: Knowledge graphs of the 25 largest banks at the common cut-off $c = 0.035$.

Figure 2 draws the 25 largest banks at the common cut-off $c = 0.035$. The crisis graph carries 244 of the 300 possible edges, and the non-crisis graph carries 175. The denser crisis graph aligns with the liquidity discount index result of Chen and Zhang (2024), who find firms more closely connected during the crisis. All banks except IBN, HDB and LYG have 17 to 24 edges and form a dense core, while these three, with 10 to 12 edges, lie on a tail at the edge. The cluster colors are scattered across that core rather than aligned with it, so the clustering does not corroborate the graph inside the crisis.

The non-crisis graph is visually sparser and more organized. It separates into a North American group of {C, BAC, BMO, TD, CM, RY, JPM, BNS and PNC} together with UBS and the two Indian listings, and a British and continental European group of {NWG, ING, BCS, LYG, HSBC, SAN and BBVA}, joined through {USB, WFC} and the Japanese and Brazilian names {MUFG, SMFG, MFG and ITUB}. HSBC and LYG are the least connected, at 9 edges

each, but HSBC–LYG carries the largest partial correlation in the graph, 0.105, and 6 of the 9 edges of HSBC link it to European names. Here, the cluster colors follow the two groups, so the clustering corroborates the graph.

Compared with the crisis graph, 4 of the 8 crisis edges above 0.050, all linking IBN and HDB to LYG and NWG, fall below the cut-off outside the crisis. Figure 3 plots the edge counts $E(c)$ against the cut-off. The two counts cross at $c = 0.040$, and below the crossing the crisis graph is denser. At $c = 0.020$ it is almost complete at 298 edges against 253. Above the crossing, the ordering reverses, and the gap peaks at $c = 0.045$, with 29 edges against 119. At $c = 0.050$ the crisis graph holds 8 edges and the non-crisis graph 84, and at $c = 0.075$ the figures are 1 and 8. The denser crisis graph therefore rests on partial correlations below 0.040 in absolute value, and the larger partial correlations belong mostly to the non-crisis subsample, where the surviving edges resolve into a North American block and a European block joined through USB, WFC and the Japanese and Brazilian names. This matches the volatility result of Chen and Zhang (2024), which they describe as counterintuitive, since one would expect financial firms to be more connected during the crisis.

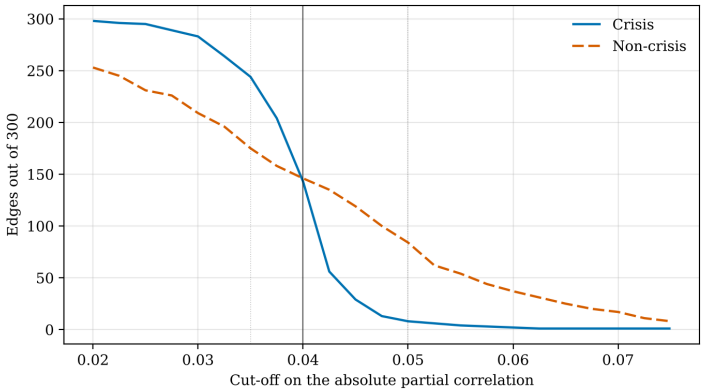


Figure 3: Edge counts $E(c)$ against the common cut-off c on the absolute partial correlation. The solid vertical line marks the crossing at $c = 0.040$ and the dotted lines mark $c = 0.035$ and $c = 0.050$.

5 Conclusion

This project examines whether the volatilities of banks listed in the United States were more closely connected during the COVID-19 crisis than outside it. Following Chen and Zhang (2024), we fit the graphical lasso to the standardized 1-year volatility of the 25 largest banks inside and outside the crisis and draw an edge where the absolute partial correlation exceeds a cut-off common to both graphs. The crisis graph is denser at small cut-offs, with 244 edges against 175 at $c = 0.035$, but the ordering reverses from $c = 0.040$, and at $c = 0.050$ the crisis graph keeps 8 edges against 84, consistent with the volatility result of Chen and Zhang (2024).

The comparison has two limitations. The 1-year window lets crisis returns enter the non-crisis volatility until May 2021, and the edge counts are point estimates without a measure of sampling uncertainty. Future research could attach sampling uncertainty to each edge and apply the liquidity discount index of Chen et al. (2016) to the 2020 episode.

References

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